TRIGONOMETRY

त्रिकोणमिति

SHEET-05

CLASS NOTES

Multiple and Sub-multiple Trigonometric function of 2A
(2A के गुणक तथा उपगुणक त्रिकोणमितीय फलन)

- $\sin 2A = 2\sin A.\cos A = \dfrac{2\tan A}{1+\tan^2 A}$
- $\sin A = 2\sin\dfrac{A}{2}\cos\dfrac{A}{2} = \dfrac{2\tan\dfrac{A}{2}}{1+\tan^2\dfrac{A}{2}}$
- $\cos 2A = \cos^2 A - \sin^2 A = 2\cos^2 A - 1$
  $= 1 - 2\sin^2 A = \dfrac{1-\tan^2 A}{1+\tan^2 A}$
- $\cos A = \cos^2\dfrac{A}{2} - \sin^2\dfrac{A}{2} = 2\cos^2\dfrac{A}{2} - 1$
  $= 1 - 2\sin^2\dfrac{A}{2} = \dfrac{1-\tan^2\dfrac{A}{2}}{1+\tan^2\dfrac{A}{2}}$

- $1 - \cos A = 2\sin^2\dfrac{A}{2}$
- $1 + \cos A = 2\cos^2\dfrac{A}{2}$
- $\tan 2A = \dfrac{2\tan A}{1-\tan^2 A}$
- $\tan A = \dfrac{2\tan\dfrac{A}{2}}{1-\tan^2\dfrac{A}{2}}$
- $\cot 2A = \dfrac{\cot^2 A - 1}{2\cot A}$

20

$$* \sin 2A = 2\sin A \cos A = \frac{2\tan\theta}{1+\tan^2\theta}$$

$$* \cos 2A = \cos^4\theta - \sin^4\theta$$
$$= \cos^2\theta - \sin^2\theta$$
$$= 1 - 2\sin^2\theta$$
$$= 2\cos^2\theta - 1$$
$$= \frac{1-\tan^2\theta}{1+\tan^2\theta}$$

$$* \tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$$

Trigonometric function of 3A
(3A के त्रिकोणमितीय फलन)

- $\sin 3A = 3\sin A - 4\sin^3 A$
- $\cos 3A = 4\cos^3 A - 3\cos A$
- $\tan 3A = \left(\dfrac{3\tan A - \tan^3 A}{1 - 3\tan^2 A}\right)$
- $\cot 3A = \left(\dfrac{\cot^3 A - 3\cot A}{3\cot^2 A - 1}\right)$

1. If $2(\cos^2\theta - \sin^2\theta) = 1$ (θ is a positive acute angle), then $\cot\theta$ is equal to :

(a) $-\sqrt{3}$ (b) $\frac{1}{\sqrt{3}}$

(c) 1 (d) $\sqrt{3}$

$2\times\cos 2\theta = 1$

$\therefore \cos 2\theta = \frac{1}{2} = \cos 60^\circ$

$2\theta = 60^\circ$

$\theta = 30^\circ$

$\cot 30^\circ = \sqrt{3}$

2. The value of

$$\left(\frac{2\cos^3\theta-\cos\theta}{\sin\theta-2\sin^3\theta}\right)^2+1, \theta \neq 45^\circ \text{ is:}$$

SSC CPO 28/06/2024 (Shift-03)

(a) $\sin^2\theta$ (b) $\sec^2\theta$

(c) $\text{cosec}^2\theta$ (d) $\cot^2\theta$

$$\left[\frac{\cos\theta(2\cos^2\theta-1)}{\sin\theta(1-2\sin^2\theta)}\right]^2+1$$

$$= \cot^2\theta+1$$

$$= \text{cosec}^2\theta$$

3. Simplify the following expression

$$\frac{\sin\theta - 2\sin^3\theta}{2\cos^3\theta - \cos\theta}$$

SSC CGL 21/07/2023 (Shift-03)

(a) $\tan\theta$ ✓

(b) $\sin\theta$

(c) $\sec\theta$

(d) $\cos\theta$

$$= \frac{S(1-2S^2)}{C(2C^2-1)}$$

$$= \tan\theta$$

Q. $\sqrt{2+\sqrt{2+2\cos 4\theta}}$

Ans $\sqrt{2+\sqrt{2(1+\cos 4\theta)}}$

$= \sqrt{2+\sqrt{2\times 2\cos^2 2\theta}}$

$= \sqrt{2+2\cos 2\theta}$

$= \sqrt{2(1+\cos 2\theta)}$

$= \sqrt{2\times 2\cos^2\theta}$

$= 2\cos\theta$

Trick: $2\cos\theta$

$1+\cos 2\theta = 2\cos^2\theta$

$1+\cos 4\theta = 2\cos^2 2\theta$

4. Evaluate the following:

$$\sqrt{2+\sqrt{2+\sqrt{2+2\cos 8\theta}}}$$

θ 2θ 4θ

SSC CGL 27/07/2023 (Shift-3)

(a) $2\cos\theta$ **(b) $2\cos 2\theta$**

(c) $\sin 2\theta$ **(d) $\cos 2\theta$**

$= 2\cos\theta$

$4\tan\theta = 3$

$\tan\theta = \frac{3}{4}$

5. If $4\tan\theta - 3 = 0$, then the value of $\frac{1-\cos 2\theta}{1+\cos 2\theta}$ is?

SSC CGL 23/09/2024 (Shift-01)

(a) 1

(b) $\frac{7}{15}$

(c) $\frac{4}{3}$

(d) $\frac{9}{16}$

$$\frac{1-(1-2\sin^2\theta)}{1+(2\cos^2\theta-1)}$$

$$= \frac{1-1+2\sin^2\theta}{1+2\cos^2\theta-1} = \tan^2\theta = \frac{9}{16}$$

6. $\frac{2\tan A}{1+\tan^2 A} = ?$

SSC CGL 20/07/2023 (Shift-02)

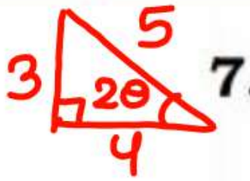
(a) cos2A

(b) sinA

(c) cosA

(d) sin2A

$\cos 2\theta = \frac{4}{5}$



7. If $\cos^4\theta - \sin^4\theta = \frac{4}{5}$, then find the value of $\sin 4\theta$.

SSC CHSL 08/07/2024 (Shift-04)

$\sin 4\theta = 2\sin 2\theta \cos 2\theta$

$= 2 \times \frac{3}{5} \times \frac{4}{5} = \frac{24}{25}$

(a) $\frac{24}{25}$ ✓

(b) $\frac{21}{25}$

(c) $\frac{16}{25}$

(d) $\frac{18}{25}$

$\sin 2\theta = 2\sin\theta\cos\theta$

$\sin 4\theta = 2\sin 2\theta \cos 2\theta$

$\sin 6\theta = 2\sin 3\theta \cos 3\theta$

8. The value of $\frac{\sin 4\theta}{(1-\cos 4\theta)}$ is:

SSC CHSL 01/06/2022 (Shift- 03)

(a) $\cot \theta$ (b) $\cot 2\theta$

(c) $\tan \theta$ (d) $\tan 2\theta$

$$\frac{\sin 4\theta}{1-\cos 4\theta} = \frac{2\sin 2\theta \cos 2\theta}{1-(1-2\sin^2 2\theta)} = \frac{2\sin 2\theta \cos 2\theta}{1-1+2\sin^2 2\theta}$$

$$= \cot 2\theta$$

$\cos 15^\circ = \frac{(\sqrt{3}+1)}{2\sqrt{2}}$

$\cos^2 15 = \frac{(\sqrt{3}+1)^2}{8}$

$= \frac{(3+1+2\sqrt{3})}{8}$

$= \frac{(4+2\sqrt{3})}{8}$

$= \left(\frac{2+\sqrt{3}}{4}\right)$

9. What is the value of $\cos^2 15^\circ$?

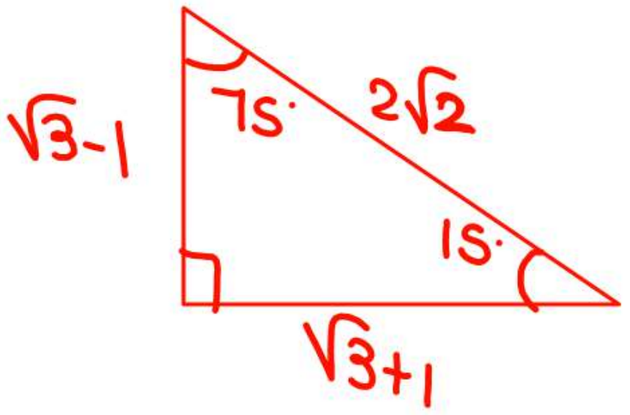
SSC CGL 06/12/2022 (Shift- 04)

(a) $(2 + \sqrt{3})$

(b) $\frac{(2+\sqrt{3})}{4}$

(c) $\frac{(2+\sqrt{3})}{2}$

(d) $\frac{(1+\sqrt{3})}{2}$



10. What is the value of cosec15° sec15°

SSC CGL 13/12/2022 (Shift- 02)

(a) 0.5 **(b) 4**

(c) 2 **(d) 1**

$$\text{ans} = \frac{2 \times \frac{1}{2\sin 15^\circ} \times \frac{1}{\cos 15^\circ}}{}$$

$$= \frac{2}{\sin 30^\circ}$$

$$= \frac{2}{\frac{1}{2}} = 4$$

11. What is the value of the expression $100(\sin 15^\circ \cos 15^\circ)$?

SSC CHSL 03/06/2022 (Shift- 03)

(a) 50 (b) 75

(c) 100 (d) 25

$100 \times \frac{2 \times \sin 15 \times \cos 15}{2}$

So

$= 100 \times \frac{\sin 30}{2}$

$= 25 \times \frac{1}{2}$

$\frac{1-\tan^2 22.5}{1+\tan^2 22.5} = \cos 45^\circ = \frac{1}{\sqrt{2}}$

12. Find the value of $\frac{1-\tan^2 22\frac{1}{2}^\circ}{1+\tan^2 22\frac{1}{2}^\circ}$ is :

(a) $\frac{\sqrt{3}}{2}$

(b) $\frac{1}{\sqrt{2}}$

(c) $\frac{1}{2}$

(d) $\sqrt{3}$

$\cos 2\theta = \frac{1-\tan^2\theta}{1+\tan^2\theta}$

2θ

* $\sin 2\theta = 2\sin\theta\cos\theta = \frac{2\tan\theta}{1+\tan^2\theta}$

* $\cos 2\theta = \cos^4\theta - \sin^4\theta$

$= \cos^2\theta - \sin^2\theta$

$= 1 - 2\sin^2\theta$

$= 2\cos^2\theta - 1$

$= \frac{1-\tan^2\theta}{1+\tan^2\theta}$

* $\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$

3θ

- $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$
- $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$

13. Find the value of $3\sin 15^\circ - 4\sin^3 15^\circ$.

SSC CGL 25/09/2024 (Shift-03)

(a) 2 **(b) $\frac{1}{2}$**

(c) $\frac{1}{\sqrt{2}}$ **(d) $\sqrt{2}$**

$= 3\sin\theta - 4\sin^3\theta$

$= \sin 3\theta$

$= \sin 45 = \frac{1}{\sqrt{2}}$

* $\sqrt{\frac{1-S}{1+S}} = \sec\theta - \tan\theta$

* $\sqrt{\frac{1+S}{1-S}} = \sec\theta + \tan\theta$

* $\sqrt{\frac{1-C}{1+C}} = \operatorname{cosec}\theta - \cot\theta$

* $\sqrt{\frac{1+C}{1-C}} = \operatorname{cosec}\theta + \cot\theta$

14. Find the value of $\sqrt{\frac{1-\sin 3\theta}{1+\sin 3\theta}}$

SSC CGL TIER I 21/07/2023 (Shift-02)

(a) $\sec 3\theta - \tan 3\theta$ (b) $(\sec 3\theta - \tan 3\theta)^2$

(c) $(\sec 3\theta - \tan 3\theta)^3$ (d) $\sec 3\theta + \tan 3\theta$

ans:- $\sec 3\theta - \tan 3\theta$

15. What is the value of $(4\sin^3 x - 3\sin x + \sin 3x)$?

SSC CHSL 31/05/2022 (Shift- 02)

(a) 1 (b) 0

(c) 4 (d) 3

$= (4S^3 - 3S) + \sin 3x$

$= -(3S - 4S^3) + \sin 3x$

$= -\sin 3x + \sin 3x$

$= 0$

$\sin\theta = \frac{\sqrt{3}}{2} = \sin 60^\circ$

$\theta = 60^\circ$

$\therefore \cos 180^\circ = -1$

16. If $\sin\theta = \frac{\sqrt{3}}{2}$, and θ is an acute angle, find the value of $\cos 3\theta$.

SSC CHSL 10/06/2022 (Shift- 01)

(a) $\frac{1}{2}$ **(b)** 1

(c) – 1 **(d)** 0

$\theta = 180^\circ$

$\sin 180^\circ = 0$

$\cos 180^\circ = -1$

$\tan 180^\circ = 0$

17. If $\sin\theta = \frac{1}{2}$, then the value of $(3\cos\theta - 4\cos^3\theta)$ is:

SSC CGL 06/12/2022 (Shift- 03)

(a) 0
(b) 1
(c) 2
(d) −1

$\sin\theta = \frac{1}{2}$

$\theta = 30^\circ$

$-(4\cos^3\theta - 3\cos\theta)$

$= -\cos 3\theta$

$= -\cos 90^\circ$

$= 0$

18. What will be the value of $\sin 10^\circ - \frac{4}{3}\sin^3 10^\circ$?

SSC CGL 06/12/2022 (Shift-01)

(a) $\frac{1}{3\sqrt{3}}$

(b) $\frac{1}{6}$

(c) $\frac{1}{2\sqrt{3}}$

(d) $\frac{\sqrt{3}}{6}$

$$\sin 10^\circ - \frac{4}{3}\sin^3 10$$

$$= \frac{3\sin 10^\circ - 4\sin^3 10^\circ}{3}$$

$$= \frac{\sin 30^\circ}{3} = \frac{1}{2\times 3} = \frac{1}{6}$$

Identities based on Sum & Difference of angle of T-Ratio
(त्रिकोणमिति अनुपात के कोणों का योग तथा अंतर पर आधारित सर्वसमिकाएँ)

- $\sin(A + B) = \sin A \cos B + \cos A \sin B$
- $\sin(A - B) = \sin A \cos B - \cos A \sin B$
- $\cos(A + B) = \cos A \cos B - \sin A \sin B$
- $\cos(A - B) = \cos A \cos B + \sin A \sin B$
- $\tan(A + B) = \dfrac{\tan A + \tan B}{1 - \tan A \tan B}$
- $\tan(A - B) = \dfrac{\tan A - \tan B}{1 + \tan A \tan B}$
- $\cot(A + B) = \dfrac{\cot A \cot B - 1}{\cot B + \cot A}$
- $\cot(A - B) = \dfrac{\cot A \cot B + 1}{\cot B - \cot A}$

19. Which of the following options gives an expression equivalent to sin(A + B)?

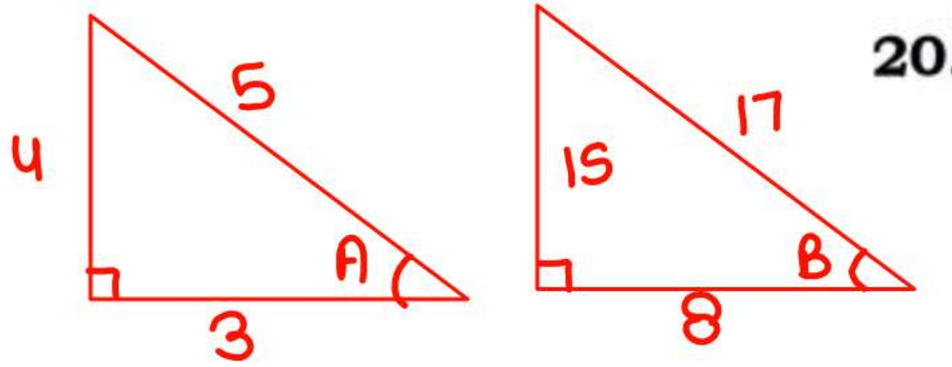
SSC CGL 24/07/2023 (Shift-2)

(a) cosAcosB – sinAsinB

(b) sinAcosB + cosAsinB

(c) cosAcoSB + sinAsinB

(d) sinAcosB-cosAsinB



20. If $\sin A = \frac{4}{5}$ and $\sin B = \frac{15}{17}$, what is the value of $\sin(A - B)$?

SSC CGL 02/12/2022 (Shift- 04)

(a) $-\frac{32}{45}$

(b) $-\frac{13}{85}$

(c) $\frac{13}{85}$

(d) $\frac{32}{45}$

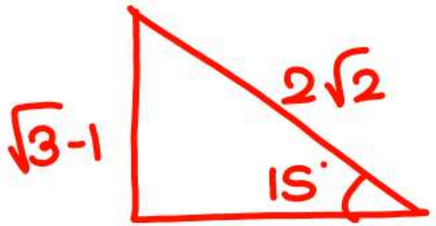
$\sin(A-B) = \sin A \cos B - \cos A \sin B$

$= \frac{4}{5} \times \frac{8}{17} - \frac{3}{5} \times \frac{15}{17}$

$= -\frac{13}{85}$

21. What is the value of cos45° sin15°?

SSC CGL 07/12/2022 (Shift- 02)



(a) $\frac{(\sqrt{3}-1)}{2}$ (b) $\frac{(\sqrt{3}-1)}{4}$

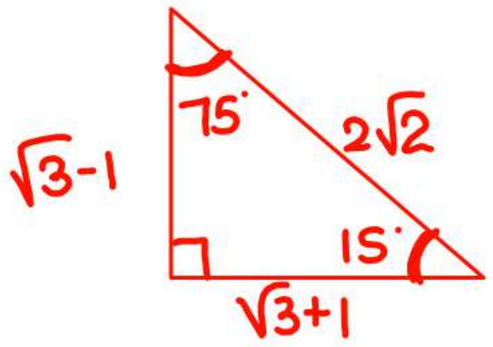
(c) $(\sqrt{3}+1)$ (d) $2\sqrt{3}-1$

$\frac{1}{\sqrt{2}} \times \frac{(\sqrt{3}-1)}{2\sqrt{2}}$

$= \frac{(\sqrt{3}-1)}{4}$

22. What is the value of sin75° + sin15°?

SSC CGL 09/12/2022 (Shift- 04)



(a) $\frac{1}{\sqrt{2}}$

(b) $\frac{\sqrt{3}}{2}$

(c) $\sqrt{\frac{3}{2}}$

(d) $\frac{3}{\sqrt{2}}$

$\sin 75 + \sin 15$

$= \frac{\sqrt{3}+1}{2\sqrt{2}} + \frac{\sqrt{3}-1}{2\sqrt{2}}$

$= \frac{2\sqrt{3}}{2\sqrt{2}}$

$= -(\cos 25 \cos 65^\circ - \sin 25 \sin 65)$

$= -(\cos 90^\circ)$

$= 0$

23. Evaluate the following

$+\sin 25^\circ \sin 65^\circ - \cos 25^\circ \cos 65^\circ$.

SSC CGL 14/07/2023 (Shift-2)

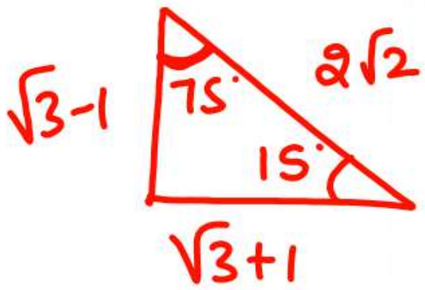
(a) 40 **(b) 4**

(c) 0 **(d) 1**

$\cos(A+B) = \cos A \cos B - \sin A \sin B$

24. Using cos (A + B) = cosA cosB – sin A sin B, find the value of cos75°

SSC CGL 17/07/2023 (Shift-03)



$\cos 75^\circ = \frac{\sqrt{3}-1}{2\sqrt{2}} = \frac{\sqrt{6}-\sqrt{2}}{4}$

(a) $\frac{\sqrt{5}-1}{4}$

(b) $\frac{\sqrt{5}+1}{4}$

(c) $\frac{\sqrt{6}-\sqrt{2}}{4}$

(d) $\frac{\sqrt{6}+\sqrt{2}}{4}$

$\cos(x+y) - \sin(x-y) + \tan 2z$

$= \cos x\cos y - \sin x\sin y - \sin x\cos y + \cos x \sin y + \frac{2\tan z}{1-\tan^2 z}$

25. What is the value of $\cos(x + y) - \sin(x - y) + \tan(2z)$?

SSC CHSL 02/07/2024 (Shift-01)

(a) $\cos x.\cos y - \sin x.\sin y - \sin x.\cos y - \cos x.\sin y + \frac{2\tan z}{1-\tan^2 z}$

(b) $\cos x.\cos y - \sin x.\sin y - \sin x.\cos y + \cos x.\sin y + \frac{2\tan z}{1-\tan^2 z}$

(c) $\cos x.\cos y - \sin x.\sin y - \sin x.\cos y + \cos x.\sin y + \frac{\tan z}{1-\tan^2 z}$

(d) $\sin x.\sin y + \frac{2\tan z}{1-\tan^2 z}$

26. Using $\tan(A - B) = \dfrac{\tan A - \tan B}{1 + \tan A \tan B}$, find the value of $\tan 15°$.

H.W

SSC CGL 18/07/2023 (Shift-04)

(a) $\sqrt{3} + 1$

(b) $\sqrt{3} - 1$

(c) $2 - \sqrt{3}$

(d) $2 + \sqrt{3}$

$\tan 30^\circ = \frac{1}{\sqrt{3}}$

27. What will be the value of $\frac{\tan 45^\circ - \tan 15^\circ}{1 + \tan 45^\circ \tan 15^\circ}$?

SSC CGL 07/12/2022 (Shift-03)

(a) $\sqrt{3}$

(b) $\frac{1}{\sqrt{2}}$

(c) $\frac{1}{\sqrt{3}}$

(d) $\sqrt{2}$

$\tan(x-y) = \frac{\tan x - \tan y}{1 + \tan x \tan y}$

$\tan(x-y) = \frac{\tan x - \tan y}{1 + \tan x \tan y}$

28. **Find the value of**

$\tan 72° - \tan 27° - \tan 72° \tan 27°$.

SSC CGL 25/09/2024 (Shift-02)

(a) –1 **(b) 1**

(c) –2 **(d) 0**

$\Rightarrow \tan(72°-27°) = \frac{\tan 72° - \tan 27°}{1 + \tan 72° \tan 27°}$ ($\tan 45°$)

$\Rightarrow 1 + \tan 72 \tan 27 = \tan 72 - \tan 27$

$\Rightarrow 1 = \tan 72 - \tan 27 - \tan 72 \cdot \tan 27$

$72° - 27° = 45°$

28. Find the value of

$\tan 72^\circ - \tan 27^\circ - \tan 72^\circ \tan 27^\circ$.

SSC CGL 25/09/2024 (Shift-02)

(a) –1 **(b) 1**

(c) –2 **(d) 0**

$\because 72^\circ - 27^\circ = 45^\circ$

$\tan 72 - \tan 27 - \tan 72 \tan 27 = 1$

If $(A-B)=45^\circ$

$\therefore \quad \tan(A-B) = \tan 45^\circ$

$\Rightarrow \dfrac{\tan A - \tan B}{1+\tan A \tan B} = 1$

$\Rightarrow \quad \tan A - \tan B = 1 + \tan A \tan B$

$\Rightarrow$ $\boxed{\tan A - \tan B - \tan A \tan B = 1}$

Ex $\tan 80^\circ - \tan 35^\circ - \tan 80^\circ \cdot \tan 35^\circ = 1$

Ex $\tan 55 - \tan 10 - \tan 55 \cdot \tan 10 = 1$

If $(A+B)=45^\circ$

$\tan(A+B) = \tan 45^\circ$

$\Rightarrow \dfrac{\tan A + \tan B}{1 - \tan A \tan B} = 1$

$\Rightarrow \quad \tan A + \tan B = 1 - \tan A \tan B$

$\Rightarrow$ $\boxed{\tan A + \tan B + \tan A \tan B = 1}$

Ex: $\tan 40 + \tan 5 + \tan 40 \cdot \tan 5 = 1$

Ex: $\tan 25 + \tan 20 + \tan 25 \cdot \tan 20 = 1$

If $A+B=45^\circ$

$\Rightarrow \boxed{B=45^\circ-A^\circ}$

$\Rightarrow \tan B = \tan(45-A)$

$\Rightarrow \tan B = \dfrac{\tan 45 - \tan A}{1+\tan 45 \tan A}$

$$\boxed{\tan B^\circ = \frac{1-\tan A}{1+\tan A}}$$

Ex:- $\dfrac{1-\tan 15}{1+\tan 15} = \tan 30^\circ = \dfrac{1}{\sqrt{3}}$

$\dfrac{1-\tan 10}{1+\tan 10} = \tan 35^\circ$

Q. $\frac{1-\tan A}{1+\tan A} = \tan(45-A)$

Q. $\frac{1+\tan A}{1-\tan A} = \tan(45+A)$

Q. $\frac{1-\tan 15}{1+\tan 15} = \tan(45-15) = \tan 30^\circ$

Q. $\frac{1+\tan 15}{1-\tan 15} = \tan(45+15) = \tan 60^\circ$

29. Simplify: $\frac{\cos 13^\circ + \sin 13^\circ}{\cos 13^\circ - \sin 13^\circ}$

SSC CPO 28/06/2024 (Shift-01)

(a) cot 58°
(b) tan 32°
(c) tan 58°
(d) cos 26°

$\frac{1+\tan 13^\circ}{1-\tan 13^\circ} = \tan(45+13) = \tan 58^\circ$

$$\tan 2\alpha$$

$$= \tan\{(A+B)+(A-B)\}$$

$$= \frac{\tan(A+B)+\tan(A-B)}{1-\tan(A+B)\tan(A-B)}$$

$$= \frac{a+b}{1-ab}$$

30. If $\tan(\alpha+\beta)=a$, $\tan(\alpha-\beta)=b$, then the value of $\tan 2\alpha$ is:

SSC CGL 13/12/2022 (Shift- 04)

(a) $\frac{a+b}{1-ab}$ ✓

(b) $\frac{a+b}{1+ab}$

(c) $\frac{a-b}{1+ab}$

(d) $\frac{a-b}{1-ab}$

31. Using $\cot(A - B) = \dfrac{\cot A \cot B + 1}{\cot B - \cot A}$, find the value of $\cot 15°$.

SSC CGL 19/07/2023 (Shift-02)

(a) $2 + \sqrt{3}$

(b) $2 - \sqrt{3}$

(c) $\sqrt{3} - 1$

(d) $\sqrt{3} + 1$

$$\tan(A+B) = 1$$

$$\therefore \tan(A+B) = 1$$

$$\Rightarrow \frac{\tan A + \tan B}{1 - \tan A \tan B} = 1$$

$$\Rightarrow \tan A + \tan B = 1 - \tan A \tan B$$

$$\Rightarrow \tan A(1 + \tan B) = 1 - \tan B$$

$$\Rightarrow \tan A = \frac{1 - \tan B}{1 + \tan B}$$

32. If $\sin(A + B) = \cos(A + B)$, what is the value of $\tan A$?

SSC CGL 02/12/2022 (Shift- 01)

(a) $\frac{1 - \tan B}{1 + \tan B}$

(b) $\frac{1 + \tan B}{1 - \tan B}$

(c) $\frac{1 + \sec B}{1 - \sec B}$

(d) $\frac{1 - \operatorname{cosec} B}{1 + \operatorname{cosec} B}$

$\tan(A+B) = 1$

$\therefore \quad \tan(A+B) = \tan 45^\circ$

$A = 45 - B$

$\tan A = \frac{1-\tan B}{1+\tan B}$

$\tan B = \frac{1-\tan A}{1+\tan A}$

32. If $\sin(A + B) = \cos(A + B)$, what is the value of $\tan A$?

SSC CGL 02/12/2022 (Shift- 01)

(a) $\frac{1 - \tan B}{1 + \tan B}$

(b) $\frac{1 + \tan B}{1 - \tan B}$

(c) $\frac{1 + \sec B}{1 - \sec B}$

(d) $\frac{1 - \operatorname{cosec} B}{1 + \operatorname{cosec} B}$

Transformation of sum or difference into product
(योग अथवा अन्तर का गुणन में परिवर्तन)

- $\sin C + \sin D = 2 \sin\left(\frac{C+D}{2}\right) \cos\left(\frac{C-D}{2}\right)$
- $\sin C - \sin D = 2 \cos\left(\frac{C+D}{2}\right) \sin\left(\frac{C-D}{2}\right)$
- $\cos C + \cos D = 2 \cos\left(\frac{C+D}{2}\right) \cos\left(\frac{C-D}{2}\right)$
- $\cos C - \cos D = 2 \sin\left(\frac{C+D}{2}\right) \sin\left(\frac{D-C}{2}\right)$

$$* \sin C + \sin D = 2 \sin \frac{(C+D)}{2} \cos \frac{(C-D)}{2}$$

$$* \sin C - \sin D = 2 \cos \frac{(C+D)}{2} \sin \frac{(C-D)}{2}$$

$$* \cos C + \cos D = 2 \cos \frac{(C+D)}{2} \cos \frac{(C-D)}{2}$$

$$* \cos C - \cos D = 2 \sin \frac{(C+D)}{2} \sin \frac{(D-C)}{2}$$

33. If $\sin C + \sin D = x$, then the value of x is :

(a) $2\sin\left(\frac{C+D}{2}\right)\sin\left(\frac{C-D}{2}\right)$

(b) $2\sin\left(\frac{C-D}{2}\right)\cos\left(\frac{C+D}{2}\right)$

(c) $2\cos\left(\frac{C+D}{2}\right)\cos\left(\frac{C-D}{2}\right)$

(d) $2\sin\left(\frac{C+D}{2}\right)\cos\left(\frac{C-D}{2}\right)$

34. What is $\sin\alpha - \sin\beta$?

SSC CGL 01/12/2022 (Shift- 02)

(a) $2\cos\frac{\alpha+\beta}{2}\sin\frac{\alpha-\beta}{2}$

(b) $2\sin\frac{\alpha+\beta}{2}\sin\frac{\alpha-\beta}{2}$

(c) $2\cos\frac{\alpha-\beta}{2}\sin\frac{\alpha+\beta}{2}$

(d) $2\cos\frac{\alpha+\beta}{2}\cos\frac{\alpha-\beta}{2}$

$2\sin\frac{(C+D)}{2}\ \sin\frac{(D-C)}{2}$

$-2\sin\frac{(C+D)}{2}\ \sin\frac{(C-D)}{2}$

35. If $\cos C - \cos D = y$, then the value of y is :

(a) $2\sin\left(\frac{C+D}{2}\right)\sin\left(\frac{C-D}{2}\right)$

(b) $2\cos\left(\frac{C+D}{2}\right)\cos\left(\frac{C-D}{2}\right)$

(c) $-2\sin\left(\frac{C+D}{2}\right)\sin\left(\frac{C-D}{2}\right)$

(d) $-2\cos\left(\frac{C+D}{2}\right)\cos\left(\frac{C-D}{2}\right)$

36. Find the value of: (sin 75° + sin 15°).

SSC Phase-XII 21/06/2024 (Shift-03)

(a) $\frac{\sqrt{6}}{2}$ (b) $\sqrt{6}$

(c) $\frac{\sqrt{3}}{2}$ (d) $\sqrt{3}$

$\sin 75^\circ + \sin 15^\circ$

$= 2 \sin 45^\circ \cos 30^\circ$

$= 2 \times \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{\sqrt{2}} = \frac{\sqrt{6}}{2}$

37. Find $\dfrac{\cos 7x + \cos 5x}{\sin 7x - \sin 5x} = ?$

(a) $\cot 2x$ (b) $\cot 3x$

(c) $\cot x$ (d) $\cot 6x$

$$= \frac{2\cos\frac{12x}{2}\cdot\cos\frac{2x}{2}}{2\cos\frac{12x}{2}\sin\frac{2x}{2}} = \cot x$$

$\cos 2x = 1 - 2\sin^2 x$

$\Rightarrow \boxed{\cos 2x - 1 = -2\sin^2 x}$

38. Find $\dfrac{\sin 5x - 2\sin 3x + \sin x}{\cos 5x - \cos x} = ?$

(a) $\tan 4x$ **(b)** $\tan 3x$

(c) $\tan 2x$ **(d)** $\tan x$

$$\frac{\sin 5x + \sin x - 2\sin 3x}{\cos 5x - \cos x}$$

$$= \frac{2\sin 3x \cdot \cos 2x - 2\sin 3x}{2\sin 3x \sin(-2x)} = \frac{2\sin 3x\,(\cos 2x - 1)}{-2\sin 3x \sin 2x}$$

$$= \frac{+2\sin^2 x}{+2\sin x \cos x} = \tan x$$

39. Master question

What is the value of

$$\frac{[(\sin x + \sin y)(\sin x - \sin y)]}{[(\cos x + \cos y)(\cos x - \cos y)]}$$ **is :**

(a) 0 **(b) 1**

(c) – 1 **(d) 2**

$$= \frac{2\sin\frac{(x+y)}{2}\cos\frac{(x-y)}{2} \times 2\cos\frac{(x+y)}{2}\sin\frac{(x-y)}{2}}{-2\cos\frac{(x+y)}{2}\cos\frac{(x-y)}{2} \times 2\sin\frac{(x+y)}{2}\sin\frac{(x-y)}{2}} = \frac{1}{-1} = -1$$

40. What is the value of $\sin 75^\circ + \sin 15^\circ$?

(a) $\sqrt{3}$

(b) $\frac{2}{\sqrt{3}}$

(c) $\sqrt{\frac{3}{2}}$

(d) $\frac{3}{\sqrt{2}}$

41. Simplify the following:

$\sin 2x + 2\sin 4x + \sin 6x$

SSC CGL 18/07/2023 (Shift-02)

(a) $4\cos^2 x \sin 4x$ (b) $4\cos^2 x \sin x$

(c) $2\cos^2 x \sin 4x$ (d) $4\sin^2 x \sin 4x$

$= \sin 6x + \sin 2x + 2\sin 4x$

$= 2\sin 4x \cos 2x + 2\sin 4x$

$= 2\sin 4x(\cos 2x + 1)$

$= 2\sin 4x \times 2\cos^2 x$

$= 4\cos^2 x \sin 4x$

Transformation of a product into sum or difference
(गुणन का योग या अन्तर में परिवर्तन)

(i) $\sin(A + B)\sin(A - B) = \sin^2 A - \sin^2 B$

$= \cos^2 B - \cos^2 A$

(ii) $\cos(A+B)\cos(A - B) = \cos^2 A - \sin^2 B$

$= \cos^2 B - \sin^2 B$

(iii) $2 \sin A \cos B = \sin(A + B) + \sin(A - B)$

(iv) $2 \cos A \sin B = \sin(A + B) - \sin(A - B)$

(v) $2 \cos A \cos B = \cos(A+B) + \cos(A - B)$

(vi) $2 \sin A \sin B = \cos(A - B) - \cos(A + B)$

* $\sin(x+y) = \sin x \cos y + \cos x \sin y$ —— (i)

* $\sin(x-y) = \sin x \cos y - \cos x \sin y$ —— (ii)

(i) + (ii)

$\sin(x+y) + \sin(x-y) = 2\sin x \cos y$.

(i) − (ii)

$\sin(x+y) - \sin(x-y) = 2\cos x \sin y$ ✗

* $\cos(x+y) = \cos x \cos y - \sin x \sin y$ —— (i)

* $\cos(x-y) = \cos x \cos y + \sin x \sin y$ —— (ii)

(i) + (ii)

$\cos(x+y) + \cos(x-y) = 2\cos x \cos y$

(ii) - (i)

$\cos(x-y) - \cos(x+y) = 2\sin x \sin y$

* $2\sin x \cos y = \sin(x+y) + \sin(x-y)$

* $2\cos x \cos y = \cos(x+y) + \cos(x-y)$

* $2\sin x \sin y = \cos(x-y) - \cos(x+y)$

$$= \frac{2\cos 75 \cdot \cos 15}{2}$$

$$= \frac{1}{2}[\cos 90 + \cos 60]$$

$$= \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

42. **Using $2\cos A \cos B = \cos(A + B) + \cos(A - B)$, find the value of $\cos 75° \cos 15°$.**

SSC CHSL 01/06/2022 (Shift- 02)

(a) $\frac{1}{2}$

(b) $\frac{1}{4}$

(c) $\frac{\sqrt{3}}{2}$

(d) $\frac{\sqrt{3}}{4}$

$= 2\sin 15^\circ \cos 45^\circ$

$= \sin 60^\circ + \sin(-30^\circ)$

$= \frac{\sqrt{3}}{2} - \frac{1}{2} = \frac{\sqrt{3}-1}{2}$

43. What will be value of $2\cos 45^\circ . \sin 15^\circ$

(a) $\frac{\sqrt{3}+1}{2}$

(b) $\frac{\sqrt{3}-2}{2}$

(c) $\frac{\sqrt{3}-1}{2}$

(d) $\frac{2}{\sqrt{3}+1}$

44. Simplify: $\dfrac{2\sin 8\theta\cos\theta - 2\sin 6\theta\cos 3\theta}{2\cos 2\theta\cos\theta - 2\sin 3\theta\sin 4\theta}$

SSC CPO 09/11/2022 (Shift-02)

(a) $\cot\theta$ **(b)** $\cot 2\theta$

(c) $\tan\theta$ **(d)** $\tan 2\theta$

$$\text{ans} = \frac{(\sin 9\theta + \sin 7\theta) - (\sin 9\theta + \sin 3\theta)}{(\cos 3\theta + \cos\theta) - (\cos\theta - \cos 7\theta)}$$

$$= \frac{\sin 7\theta - \sin 3\theta}{\cos 3\theta + \cos 7\theta} = \frac{2\cos 5\theta \sin 2\theta}{2\cos 5\theta \cos 2\theta} = \tan 2\theta$$

45. Evaluate the following:

$\cos(36^\circ + A).\cos(36^\circ - A) + \cos(54^\circ + A).\cos(54^\circ - A)$

SSC CGL 14/07/2023 (Shift-2)

(a) sin 2A (b) cos A

(c) sin A (d) cos 2A

$\cos(36+A)\cos(36-A) + \sin(36-A)\sin(36+A)$

$= \cos\{(36+A)-(36-A)\}$

$= \cos\{36+A-36+A\}$

$= \cos 2A$

$$\boxed{A+B=C}$$

$$\tan(A+B) = \tan C$$

$$\Rightarrow \quad \frac{\tan A+\tan B}{1-\tan A\tan B} = \tan C$$

$$\Rightarrow \quad \tan A+\tan B = \tan C - \tan A\tan B\tan C$$

$$\Rightarrow$$

$$\boxed{\tan A\tan B\tan C = \tan C - \tan A - \tan B}$$

46. If A + B = C, then tanA tanB tanC = ?

SSC CGL 08/12/2022 (Shift- 01)

(a) tanC + tanA – tanB

(b) tanC + tanA + tanB

(c) tanA – tanB – tanC

(d) tanC – tanA – tanB

Q. $\tan 10^\circ \tan 20^\circ \tan 30^\circ = ?$

$10+20=30^\circ$

(a) $\tan 10 - \tan 20 - \tan 30$

(b) $\tan 20 - \tan 10 - \tan 30$

(c) $\tan 10 + \tan 20 - \tan 30$

(d) ✓ $\tan 30^\circ - \tan 10 - \tan 20$

47. What is the value of the expression $\cos 2A \cos 2B + \sin^2(A - B) - \sin^2(A + B)$?

SSC CGL 12/12/2022 (Shift- 01)

(a) $\sin(2A - 2B)$ **(b) $\sin(2A + 2B)$**

(c) $\cos(2A + 2B)$ **(d) $\cos(2A - 2B)$**

$= -1$ $120 + 60$ $60^\circ = \frac{1}{2}$

$\cos 2A \cdot \cos 2B + \sin^2(A-B) - \sin^2(A+B)$

$A = 60$ $B = 30$

$= \cos 120^\circ \cos 60 + \sin^2 30 - \sin^2 90^\circ$

$= -\frac{1}{2} \times \frac{1}{2} + \frac{1}{4} - 1$

$= -1$

48. If $\cot 75° = 2 - \sqrt{3}$. Find the value of $\cot 15°$.

Try.

SSC CHSL 24/05/2022 (Shift- 03)

(a) $2 - \sqrt{3}$

(b) $2 + \sqrt{3}$

(c) $\sqrt{3} + 1$

(d) $\sqrt{3} - 1$

⊘. $A+B=45^\circ$ or 225°

Ans $(1+\tan A)(1+\tan B)=2$

$(1-\cot A)(1-\cot B)=2$

49. The value of $(1 + \tan 10°)(1 + \tan 35°)$ is:

SSC CHSL 25/05/2022 (Shift- 03)

(a) $\frac{1}{2}$ (b) $\frac{3}{4}$

(c) 1 (d) 2

$\alpha+\beta=45^\circ$

$22.5+x-y+22.5+y-x$

50. $\left[1+\tan\left(22\frac{1}{2}^\circ+x-y\right)\right]\left[1+\tan\left(22\frac{1}{2}^\circ+y-x\right)\right]$

α β

(a) 0 **(b)** 1

(c) 2 **(d)** 4

$(1+\tan\alpha)(1+\tan\beta)$

$=2$

51. Find the value of (1 + tan11°) (1 + tan 21°) (1 + tan 24°) (1 + tan 34°) (1 + tan 45°)

(a) 0 **(b) 1**

(c) 2 **(d) 8**

$11+34=45°$

$(1+\tan 11°)(1+\tan 21)(1+\tan 24°)(1+\tan 34)(1+\tan 45)$

$= 2 \times 2 \times 2$

$= 8$

Some Important Results
(कुछ महत्वपूर्ण परिणाम)

(i) $\sin A . \sin 2A . \sin 4A = \frac{1}{4} \sin 3A$

(ii) $\cos A . \cos 2A . \cos 4A = \frac{1}{4} \cos 3A$

(iii) $\tan A . \tan 2A \tan 4A = \tan 3A$

(iv) $\sin(60° - A) . \sin A . \sin(60° + A) = \frac{1}{4} \sin 3A$

(v) $\cos(60° - A) . \cos A . \cos(60° + A) = \frac{1}{4} \cos 3A$

(vi) $\tan(60° - A) . \tan A . \tan(60° + A) = \tan 3A$

(vii) $\sin^2(60° - A) + \sin^2 A + \sin^2(60° + A) = \frac{3}{2}$

(viii) $\cos^2(60° - A) + \cos^2 A + \cos^2(60° + A) = \frac{3}{2}$

* $\sin\theta \cdot \sin 2\theta \cdot \sin 4\theta = \frac{1}{4}\sin 3\theta$
* $\cos\theta \cos 2\theta \cos 4\theta = \frac{1}{4}\cos 3\theta$
* $\tan\theta \tan 2\theta \tan 4\theta = \tan 3\theta$

* $\sin\theta \cdot \sin(60-\theta) \sin(60+\theta) = \frac{1}{4}\sin 3\theta$
* $\cos\theta \cos(60-\theta) \cos(60+\theta) = \frac{1}{4}\cos 3\theta$
* $\tan\theta \tan(60-\theta) \tan(60+\theta) = \tan 3\theta$

52. Find $\sin 20^\circ . \sin 40^\circ . \sin 60^\circ . \sin 80^\circ$.

(a) $\frac{1}{16}$ (b) $\frac{5}{16}$

(c) $\frac{3}{16}$ (d) $\frac{1}{8}$

$= \sin 20 \sin 40 \sin 80 \cdot \sin 60$

$= \frac{1}{4} \sin 60 \times \sin 60$

$= \frac{1}{4} \times \frac{3}{4} = \frac{3}{16}$

53. The value of $(\tan 20^\circ . \tan 40^\circ . \tan 80^\circ)$ is :

(a) 1 (b) 0

(c) $\sqrt{3}$ (d) 3

$= \tan 60^\circ$

$= \sqrt{3}$

54. Find 1 – sin10°.sin50°.sin70°.

θ (60-θ) (60+θ)

(a) $\frac{5}{8}$ **(b)** $\frac{7}{8}$

(c) $\frac{3}{8}$ **(d)** $\frac{1}{8}$

$= 1 - \frac{1}{4}\sin 30°$

$= 1 - \frac{1}{4} \times \frac{1}{2} = 1 - \frac{1}{8} = \frac{7}{8}$

$\sin 6 \cdot \frac{\sin 54}{\sin 54} \cdot \sin 66 \sin 42 \sin 78$

$= \frac{1}{4} \frac{\sin 18 \cdot \sin 42 \sin 78}{\sin 54}$

$= \frac{1}{4} \times \frac{1}{\sin 54} \times \frac{1}{4} \sin 54 = \frac{1}{16}$

55. $\sin 6^\circ \sin 42^\circ \sin 66^\circ \sin 78^\circ = ?$

(a) $\frac{1}{4}$ (b) $\frac{1}{2}$

(c) $\frac{1}{8}$ (d) $\frac{1}{16}$

θ $60-\theta$

56. Find $\sin 12^\circ . \sin 48^\circ . \sin 54^\circ$.

$\sin 12 \sin 48 \sin 72 . \sin 54$

$\sin 72$

$= \frac{1}{4} \frac{\sin 36 \cdot \sin 54}{\sin 36 \times 2 \times \cos 36}$

$= \frac{1}{8} \times \frac{\cos 36}{\cos 36}$

(a) $\frac{5}{8}$ (b) $\frac{7}{8}$

(c) $\frac{3}{8}$ (d) $\frac{1}{8}$

57. The value of ($\cos 15^\circ . \cos 45^\circ . \cos 75^\circ$) is :

(a) $\frac{1}{3\sqrt{2}}$

(b) $\frac{1}{4\sqrt{2}}$

(c) $\frac{1}{8}$

(d) $\frac{\sqrt{3}}{8}$

$= \frac{1}{4}\cos 45^\circ$

$= \frac{1}{4} \times \frac{1}{\sqrt{2}}$

58. $\cos 10^\circ . \cos 30^\circ . \cos 50^\circ . \cos 70^\circ = ?$

(θ, $60-\theta$, $60+\theta$)

(a) $\frac{1}{16}$ (b) $\frac{3}{16}$

(c) $\frac{5}{16}$ (d) $\frac{1}{8}$

$= \frac{1}{4}\cos 30^\circ \times \cos 30^\circ$

$= \frac{1}{4} \times \frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2} = \frac{3}{16}$

$$\frac{\cos 18}{\cos 18} \cdot \cos 42 \cos 78 \cos 36^\circ$$

$$= \frac{1}{4} \frac{\cos 54 \cdot \cos 36^\circ}{\cos 18^\circ}$$

$$= \frac{1}{4} \times \frac{2}{2} \times \frac{\sin 36 \times \cos 36^\circ}{\cos 18^\circ}$$

$$= \frac{1}{8} \times \frac{\sin 72^\circ}{\cos 18^\circ}$$

59. $\cos 42^\circ . \cos 78^\circ . \cos 36^\circ = ?$

(a) $\frac{1}{2}$ (b) $\frac{1}{4}$

(c) $\frac{1}{8}$ (d) $\frac{1}{16}$

60. The value of $\tan 6^\circ . \tan 42^\circ . \tan 66^\circ . \tan 78^\circ$ is :

Try

(a) 1

(b) $\frac{1}{2}$

(c) $\frac{1}{4}$

(d) $\frac{1}{8}$

Live
- Trigonometry → Pre + Mains

VOD
- Maxima & Minima →

Live
- Height & Distance → Mains